

Intellectual Smart Hybrid AI System for Real-Time Threat Intelligence in the Internet

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Abstract— Traditional security systems where the detectors rely on signature-based sensors or independent machine learning models do not recognize these changing threats real-time and also produce too many false alerts. Here, the project is a proposal of IS-HAITI, an Intellectual Smart Hybrid AI system to be deployed in real-time cyber threat intelligence. The system combines Graph Neural Networks (GNNs) to identify relationships between nodes as a form of attack, Transformers models to analyze sequential and temporal patterns of behavior, and rule-based systems to detect previously known patterns of attacks. In comparison with the traditional systems, the IS-HAITI encourages multi-Palmar inputs (threats) such as network traffic, logs, URLs, text, images, and multi-media. The system is designed to enhance the accuracy of detection, false positives, and offer transparent and actionable insights into security using structural, temporal, rule-based intelligence along with an explainability layer to be applicable in real- world settings.

Keywords—Cybersecurity; Real-Time Threat Intelligence; Hybrid Artificial Intelligence; Graph Neural Networks; Transformer Models; Explainable AI; Multi-Model Threat Detection.

I. INTRODUCTION

Contemporary cyberattacks are no longer solitary and static, but are most often organized, multi-phase and proceeding with the evolution of the approach to go around conventional security schemes. Such methods like advanced persistent threats, polymorphic malware, automated e-mails, and insider-based attacks are a major threat to the current cybersecurity mechanisms. The conventional security controls such as firewalls, Intrusion Detection Systems (IDS), and Intrusion Prevention Systems (IPS) mainly rely on preconfigured signatures/hard coded rules. Although the techniques are useful in countering the established attack patterns, they fail miserably against zero-day exploits and other behavior patterns that have never been encountered before. In addition, most traditional systems result in large number of false alerts, which overburden the analysts at the expense of incident response.

Recent trends on Artificial Intelligence (AI) and Machine Learning (ML) led to areas where threat detection is enhanced by learning multifaceted patterns on data. Nevertheless, most of the currently available AI-based security products are based on one framework or data format and do not implement the capacity to adapt dynamically to changing attack patterns. Moreover, opaque masks of most deep learning models make them difficult to use in practice because security analysts need transparent and interpretable results that enable them to make decisions in real-life settings.

The suggested framework incorporates Graph Neural Networks (GNNs) to study structural relations among network communications, Transformer-based models to learn attack patterns in time and sequence, and rule-based operations which effectively identify known threats. These complimentary measures will strive to attain better detection rates, lower rates of false positives, and greater flexibility to new cyber threats.

IS-HAITI is engineered such that it will provide multi-model threat intelligence, which indeed involves processing a wide array of sources of security information using a common platform. A component of explainability is also included to give a clear rationale of a detection observation to allow an analyst to comprehend and justify system decision making. The suggested system is projected as a decision support system that supplements, but does not substitute human expertise thus it would be useful in implementation in real security operations.

A. Motivation and Problem Definition

The current problems in cyber security have posed major

Aspect	Traditional Security Systems	Existing AI-Based Systems	Proposed IS-HAITI Framework
Detection Basis	Predefined signatures and static rules	Learned patterns from historical data	Hybrid learning using structural, temporal, and rule-based intelligence
Zero-Day Attack Detection	Not supported	Limited, depends on training data	Explicitly designed to detect evolving and unseen attacks
Adaptability to Evolving Threats	Low (manual rule updates required)	Moderate (model retraining needed)	High (hybrid models with continuous learning capability)
Attack Coverage	Known and well-defined attack patterns	Partial (often focused on a single attack class or data type)	Broad coverage including multi-stage, coordinated, and stealthy attacks
Data Modelities	Mostly network packets or logs	Usually single-modality (traffic or logs)	Multi-model (network traffic, logs, and contextual data)

Table 1. Comparison of Traditional, Existing AI-Based, and IS-HAITI Cybersecurity Approaches

Threats to the cybersecurity systems that have developed over the years due to the increasing pace of cyber threats. According to recent research, current attacks are becoming more sophisticated, adaptive and not easily detected through the conventional ways. Advanced persistent threat, polymorphic malware, insider abuse, and multi-stage intrusion are some of the techniques currently used by attackers to avoid quantity-based signature detection and static rule-based systems. As it is being emphasized in the recent research, the traditional security mechanisms cannot detect new and cleverly concealed attacks, when the malicious activities are distributed in time and throughout various parts of the network.

A number of machine learning and deep learning methods have been suggested in order to deal with these challenges. Transformer-based models have proven to be very effective at detecting rare and suspicious network traffic patterns whereas Graph Neural Networks (GNNs) have been demonstrated to be effective at modeling intricate relations between network entities and finding orchestrating attacks. However, at the moment, solutions tend to have the theme of orientation towards a particular line of analysis, i.e., the temporal behavior, or structural relationships, which limits their abilities to reflect the entire picture of the contemporary cyber threats. Research also shows that single methodology system can be effective under controlled conditions but cannot be efficient in varied and realistic attack situations.

This work is motivated by the desire to fill these specified intention gaps by designing a coherent and hybrid AI-driven threat intelligence system, which incorporates a structural, temporal, and rule-based analysis into a single system. The main issue, which this study addresses is the lack real-time, Explainable, and flexible system of cybersecurity that can efficiently counter the emerging threats and reduce the number of false alarms and provide informed analyst decision-making.

The proposed system will fill the gap between practical usability and high detection accuracy of AI in real-world applications of cybersecurity by combining complementary AI methods and focusing on the interpretability aspect.

B. Objectives of the Proposed System

The main task of the study is to design and draft an overall and smart system of cyber threat intelligence that will be able to track and identify a wide range of cyberattacks in real time. Recent research has shown that cybersecurity systems that concentrate on a small number of models or categories of attacks may not be able to generalize to different and changing types of threats. To overcome this drawback, the proposed IS-HAITI framework will be structured to accommodate various detection models that are trained on various types of cyberattacks as opposed to limiting itself to a few or fixed techniques. The first goal of the suggested system is to facilitate the detection of various forms of attacks, such as but not limited to denial-of-service attacks, malware-based attacks, insider threats, network abnormal activities, and multi-stage attacks.

To this end, the system considers and trains several AI models, each trained to have particular attributes of various attack types. Graph Neural network based structural learning is used to identify relationship-based and coordinated attacks, whereas Transformer-based models are proposed to identify temporal and sequential activities.

The other target is the enhancement of detection reliability with low rates of false positives. Through the use of an array of trained models and integrating their solutions through a hybrid decision fusion system, the system will generate more rational threat predictions, under various attack conditions.

C. Significance of the Proposed Approach

The novelty of the suggested IS-HAITI framework is that it is

flexible, scalable, and has a broad coverage of threats. As opposed to numerous current cybersecurity solutions, which have been developed to target a dedicated range of attack types, or can use only one detection model, the proposed solution is built to support various trained models that would target various kinds of cyber threats.

The system can learn many different attack behaviors, which can be characterized by various threat features which cannot be effectively modeled using only one analytical method. Structural models do identify coordinated and graph-based patterns of attack, temporal models do identify slow and multi-stage patterns of attack, and a variety of other models may be trained to confront a specific threat, malware evolution, insider abuse, or denial-of-service attacks. The multi-model design enhances detection by a great deal but it also makes it very robust and minimizes the chances of overlooked attack.

The other vital feature of the proposed approach is that it is both explainable and practically useful. Although, in recent studies, deep learning models have shown a substantial level of detection, most systems based on deep learning are hard to interpret. An explainability aspect is needed to make the decision made during the detection transparent and actionable, which is critical to win over analyst confidence and effective incident response.

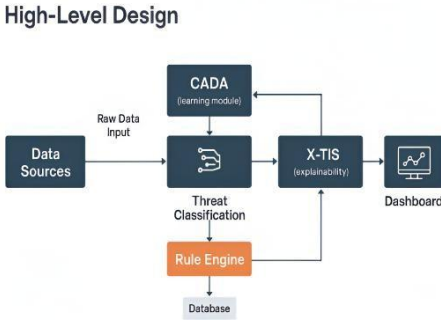


Fig 1. High-Level Architecture Framework

II. Background and Related Work

The fast-moving trend in adoption of cloud computing, Internet of things (IoT) ecosystems and utilisation of data-driven digital services have completely transformed modern cyber infrastructures. Despite the fact that these improvements provide and permit scalability and automation the result is an increase in attack surface and complicated systems. The contemporary cyber threats are not that mundane anymore but instead smart, multi-level and flexible and therefore some of them are coded in a way that they will not be detected in the long run.

Recent studies highlight the rise in the use of AI-related solutions to attacks, including automated phishing and polymorphic malware, abuse by insiders and multi-vector attacks. These types of attacks are in a continuous change of their activity that they can get away with such that the conventional static defenses are even more useless. Because of this, a paradigm

shift in the research community on cybersecurity in which the central goal has been intelligent, learning-based detection models, capable of reasoning on structure, time and heterogeneous data streams.

A. Limitations of Traditional Cybersecurity Systems

Conventional methods of cybersecurity, such as a firewall, Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS), are largely grounded on the concept of signature-based matching and already-known rules. Although efficient in identification of familiar threats, they have low adaptation, and cannot identify the zero-day exploits and new attack variations absent the manual updates.

B. AI-Based Threat Detection Systems

The vulnerability of rule-based security solutions has been solved through the introduction of artificial intelligence and machine learning techniques that are used in overcoming rigidity. The latest studies prove that supervised, unsupervised, and deep learning models are able to discover implicit trends in the big data of security data and adjust themselves to changing threats.

Deep neural models and transformer-based models have demonstrated robustness with regard to the detection of anomalies in network traffic, system logs, and malware traces. But often current AI-based systems are based on single-model architectures or unimodal data, resulting in a failure to provide the entire picture of more complex attacks. A number of researches observe that these models are sufficiently sensitive to give precise results when it comes to controlled settings, but when using more complex threat coordination or cross-domain detection, results decline, pushing the use of hybrid and multi-viewed detection methods.

C. Graph Neural Networks for Structural Threat Analysis

Graph Neural Networks (GNNs) proved beneficial in modelling cybersecurity data as relational graphs, where nodes are entities and edges are communication/interaction patterns. This representation can be used in the detection of lateral movement, coordinated intrusions and concealed communication channel, which can hardly be detected using traditional feature based methods.

More recent literature proves that GNNs are much more accurate to detect in a dense and noisy network space. Nevertheless, GNNs are mainly structured dependencies and do not have internal structures to provide long-term temporal behavior. Consequently, they are not enough in isolation in terms of detecting slow or persistent attacks.

D. Transformer Models for Temporal Behavior Detection

Transformer based models have been shown to be quite successful at learning long-range temporal association in sequential data. Their attention mechanisms support the detection of minor behavioral states over a longtime running window that makes them especially effective in detecting Advanced Persistent Threats (APTs) and insider-driven anomalies. Other recent frameworks based on transformers have been reported to be having high detection rates in all network traffic, logs of the IoT as well as process execution

traces. However, their calculation intensity is difficult to implement in real time particularly in limited or resource intensive systems. This is a factor that has seen researchers consider hybrid designs, where transformers are only applied selectively and much lighter weight models are used.

E. Hybrid AI and Rule-Based Detection Approaches

The fallacy of hybrid detection models is that it refuses to adhere to rules, but can adapt dynamically to avoid failures and embraces the responsive model. Rule engines are also effective at quickly identifying established patterns of attack, whereas learning-based components are oriented towards detecting anomalous or never-before-seen behavior.

According to the recent studies, these hybrid systems have the ability to decrease the false positives, increase stability in detection, and strengthen resistance to changing threats. Hybrid combinations of complementary detection paradigms overcome limitations of single systems.

F. Multimodel Threat Detection Systems

Contemporary cyber threats are growing in an increasing number of data modalities such as network traffic, emails, URLs, executable code files, images, audio, and video files. Multimodel threat detection systems are intended to match these dissimilar data so as to obtain an overall threat awareness.

G. Explainable Artificial Intelligence in Cybersecurity

Due to the increasing complexity of the AI-based security systems, explainability has become a key factor in functional trust and acceptance. Security analysts need to know the reason why one of the models is sent an alert so that they can legitimize their decisions and act.

Recent explainable AI (XAI) methods use visualization of attention, feature attribution and graph-based explanations to enhance transparency. Empirical research indicates that XAI-improved systems can boost analyst confidence and quality of decision making although this comes at the expense of extra computational load.

H. Continuous Learning and Adaptive Security Models

The detection models that rely on statistics do not work over time because of the changing tactics of attackers. Incremental and continuous learning methods allow models to adapt with the addition of new data and feedback without having to be retrained completely.

Federated learning has become a privacy-aware developing that can enable organizations to gain access to more valuable detection models without necessarily providing raw data. Recent work also notes that it has shown effectiveness in distributed environment besides mentioning the challenges in communication overhead and heterogeneity of models.

III. METHODOLOGY

This research proposes Intellectual Smart Hybrid AI System proposal, which is the site of the Real-Time Threat Intelligence (IS-HAITI). In contrast to the traditional unidimensional single-view detection models, IS- HAITI incorporates structural learning on the basis of Graph Neural Networks (GNNs), time-series sequence learning on the basis of Transformer architectures, and deterministic rule-based reasoning in a single unified decision making process. The suggested methodology should be capable of identifying the cyber threats known and also those unfamiliar to it, in addition to providing real-time responsiveness and explainability.

A. System Architecture

The framework suggested works on a parallel processing pipeline in order to cover all threats fully. The four major phases that make up the architecture are:

1. Data Ingestion and Feature Engineering
2. Graph Topology Construction
3. Hybrid Inference Engine
4. Risk Stratification and Alert Generation

The system processes a set of network flows

$$F=\{f1,f2,...,fn\}$$

and maps network entities into a probabilistic risk space

$$P(y | X) \in [0, 1]$$

representing the likelihood of malicious behavior.

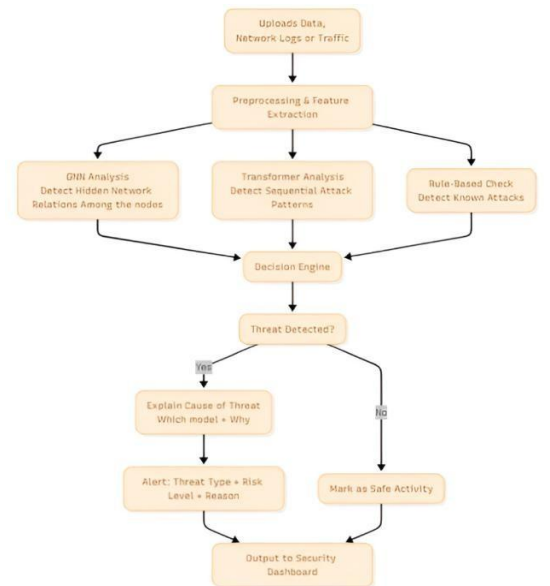


Fig 2. System Workflow Design

B. Feature Engineering and Normalization

Raw network traffic in PCAP and CSV formats is decoded to flow level characteristics. A behavioral features vector of each service node.

$$\mathbf{X}_i \in \mathbb{R}^{12}$$

is constructed to represent its activity profile.

The extracted feature set includes:

- Statistical aggregations (e.g., flow count, total bytes),
- Temporal characteristics (e.g., mean duration, burst rate),
- Protocol distribution metrics (e.g., TCP/UDP ratios)

To ensure numerical stability during gradient-based optimization, all features are normalized using Min-Max scaling:

$$x'_i = \frac{x_i - \min(x)}{\max(x) - \min(x)}$$

where x_i denotes the original feature value.

Due to the inherent imbalance in intrusion datasets (initial benign-to-attack ratio of approximately 90:10), the Synthetic Minority Over-sampling Technique (SMOTE) is applied during training to improve model sensitivity to rare attack instances.

B. Graph Construction

To capture lateral movement, coordinated behavior, and hidden attack correlations, the network is modeled as a directed graph.

$$\mathbf{G} = (\mathbf{V}, \mathbf{E})$$

To capture lateral movement, coordinated behavior, and hidden attack correlations, the network is modeled as a directed graph

- **Nodes (V)** represent unique IP addresses or network services (e.g., HTTP, SSH).
- **Edges (E)** represent communication flows, where a directed edge e_{ij} exists if node v_i initiates communication with node v_j .
- **Edge Weights (W)** are defined by the frequency of interaction between connected nodes.

This topological representation allows the GNN module to propagate behavioral information across the network, enabling each node to be evaluated in the context of its neighbors rather than in isolation.

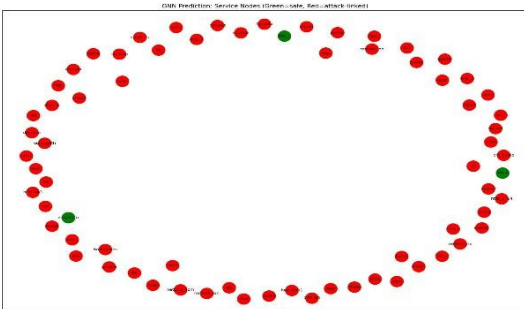


Fig 3. GNN Predicted Nodes – Red: Attack-linked, Green: Benign

C. Hybrid Detection Engine

The IS-HAITI fundamental innovation is about its hybrid inference engine, a combination of three complementary.

1) Structural Learning

A hybrid GNN architecture is employed, combining:

- **Graph Convolutional Networks (GCN)** for local neighborhood aggregation, and
- **Graph Attention Networks (GAT)** for learning weighted neighbor importance.

The model is optimized using **Binary Cross-Entropy Loss**:

$$\mathcal{L} = -N \mathbb{I} = 1 \sum N [y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i)]$$

where y_i denotes the true label and $\hat{y}_i$ represents the predicted attack probability.

2) Temporal Learning (Transformer)

In parallel, a Transformer model analyzes network flows ordered by timestamps

$$\mathbf{T} = \{t_1, t_2, \dots, t_k\}$$

with self-attention mechanisms, headlong temporal dependencies are captivated with. This allows it to identify multi-stage and low-and-slow attacks like Advanced Persistent Threats (APTs) that cannot be detected by snapshot-based detection methods.

3) Deterministic Logic (Rule-Based Engine)

A lightweight rule-based engine is used to filter known attack signatures including PINL lists of IP addresses and canned payloads. This aspect makes known threats be instantly detected with a zero-learning latency

D. Risk Scoring Algorithm

Rather than relying on a single classifier output, IS-HAITI computes an Integrated Risk Score R_v for each node v , combining probabilistic confidence with network impact:

$$R_v = \alpha \cdot P_{GNN}(v) + \beta \cdot C(v) + \gamma \cdot |S_{cluster}|$$

Where:

- $P_{GNN}(v)$ is the attack probability predicted by the GNN,
- $C(v)$ represents the network centrality of node v ,
- $|S_{cluster}|$ denotes the size of the connected compromised subgraph.

The weighting parameters are empirically set to:

$$\alpha=0.5, \beta=0.3, \gamma=0.2$$

to balance detection confidence, propagation risk, and attack scope.

V. RESULT ANALYSIS

The effectiveness of the suggested IS-HAITI framework is considered in this section based on the graph-driven detection model built on the KDDTrain+ / KDD99 traffic corpus. This is aimed at proving the benefit of structural learning over conventional flat classifiers in identifying malicious services in network traffic.

A. Experimental Setup

To test the proposed method, raw network connection records were summarized on the service level, which gave a small graph of 70 service nodes (e.g., ftp, http, irc, domain).

Graph Structure:

The nodes correspond to a service in the network and the edges to the relatedness of the services based on similarity of common neighborhoods of the protocols. Such representation allows the learning model to utilise contextual dependencies among services instead of viewing them as independent entities.

Feature Set:

In order to precisely prevent target leakage, an artificially restricted no-leak set of features was selected, namely, three aggregate measures:

- **src_bytes_mean:** Mean source bytes per service.
- **dst_bytes_mean:** Mean destination bytes per service.
- **count:** Number of aggregated connections per service.
- **Validation Protocol:** We employed 5-fold stratified cross-validation (CV) to ensure robust performance estimation, alongside a held-out validation split.

B. Performance Comparison (Cross-Validation)

Table II shows the comparison of the performance using 5-fold stratified cross-validation. Our-GNN model developed by the researchers performs better on AUC (0.962) and Precision (1.000) in comparison to traditional-baselines.

Model	Accuracy	Precision	Recall	F1-Score	AUC
Our-GNN (Proposed)	0.929	1.000	0.924	0.961	0.962
Random Forest	0.957	0.970	0.985	0.977	0.748
SVM-RBF	0.943	0.943	1.000	0.971	0.784
KNN	0.943	0.943	1.000	0.971	0.794
Logistic Regression	0.943	0.943	1.000	0.971	0.742

Table 2. PERFORMANCE COMPARISON USING 5-FOLD STRATIFIED CROSS-VALIDATION

The suggested GNN has the best AUC (0.962) and optimal percentage of accuracy (1.000) that indicates that it is very discriminative and has very few false positives and can be useful in a real-life security system. The Random Forest classifier has a slight higher raw accuracy (0.957) but a very low AUC (0.748) indicating it tends to be bad at varying decision levels.

C. Interpretation and Context-Aware Detection.

The hypothesis of context-aware detection is proven by the fact that the AUC of the GNN formulation is superior. In contrast to the baseline models (e.g. SVM, NaiveBayes) each service is assessed separately, the IS-HAITI GNN spreads risk indicators across the graph. The risk score of a service is affected by more than just the traffic statistics of the service (e.g. a large dstbytes). The behavior of the protocol-neighborhood peers also affect the risk score. This is especially successful in detecting multi related services complex attacks.

D. Explainability Analysis

We used SHAP analysis in high-risk predictions to confirm the model in predicting the choices. It is a confirmed fact that dstbytes_mean (Destination Bytes) is the most powerful feature that drives detection(Fig. 3). This is compatible with the no-leak feature set design, which ensures the model is indeed learning the anomaly in the volumes of the traffic, which is not the recall of the selected IP addresses.

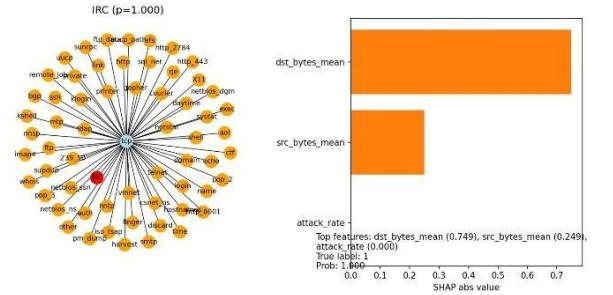


Fig 4. Explainability analysis using SHAP

The model identifies dst_bytes_mean as the primary risk factor, consistent with the feature set.

E. Receiver Operating Characteristic (ROC) Analysis

In order to test further the discriminative ability of the suggested IS-HAITI framework, we examine the Receiver Operating Characteristic (ROC) curves based on 5-fold stratified cross-validation utilizing out-of-fold predictions. The use of ROC analysis is especially significant in cybersecurity context, since it will not only evaluate the performance of various models at different decision levels, but also give an understanding of the trade off between the true positive rate (TPR) and the false positive rate (FPR).

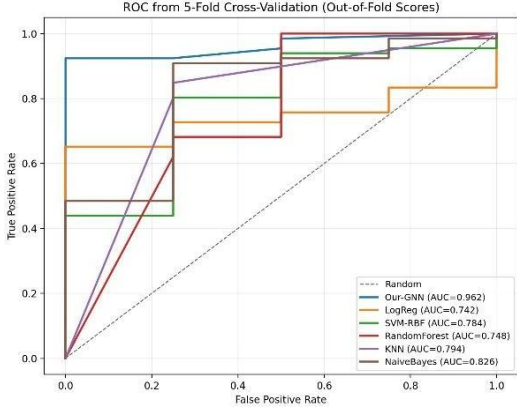


Fig 5. ROC Curve from 5-Fold Cross-Validation

As shown in Fig.5, the proposed Our-GNN model always leads in the ROC area, with the highest Area Under the Curve (AUC) of 0.962. This means that the separability between benign and malicious services is very high for all possible thresholds. In contrast, traditional classifiers like Logistic Regression (AUC = 0.742) and Random Forest (AUC = 0.748) have relatively lower separability, especially in the low false positive rate area, which is of high importance for operational security systems.

Although some traditional classifiers (like KNN and Naive Bayes) have relatively higher AUC values, their corresponding ROC curves are less stable, with a sharper decline in performance as the false positive rate increases. The robustness of the ROC curve for the GNN model clearly illustrates the benefit of structural learning, where risk information is diffused over service graphs based not only on individual feature values but also on their graph structural relationships.

F. Limitations

This experiment operates on a condensed service-aggregated graph (70 nodes) with a prevalent class imbalance. While the Cross-Validation AUC of 0.962 is robust, the perfect scores in the held-out split may reflect the limited size of the validation set. Future work will extend this evaluation to larger, time-based splits to further verify generalization.

V. CONCLUSION

The high speed of cyber threats development has revealed significant flaws in conventional protective mechanisms and Remote AI-based detection frameworks. Fixed rule-based systems are becoming ineffective to zero-day exploits, coordinated attacks, and to low-and-slow attack strategies whereas single model classifiers do not tend to be decoupled

across heterogeneous and evolving attack environments. To address these problems, the paper introduced IS-HAIT, an Intellectual Smart Hybrid AI System of Real-Time Threat Intelligence, which would offer real-time, comprehensive, and explainable cyber threats surveillance.

The proposed system combines three supplementary detectors models, including graph-based structural learning models, transformer-based temporal sequence models, and deterministic rule-based logic models, into an integrated decision system. Through an activity in the form of a network as a directed graph, IS-HAIT accommodates a set of relational and topological dependencies that are critical in revealing coordinated attack like lateral movement and botnet-like behaviour. Simultaneously, transformer-based temporal model makes it possible to detect multi-stage and persistent threats based on learning long-range dependencies in sequences of network flows, whereas the rule-based engine quickens and dependable detection of existing attack signatures.

One of the most useful contributions of the work is the hybrid inference and risk stratification method, according to which the results of all three models are combined to create a unified risk measure that incorporates the confidence of a detection and the possible impact on the network. Such a design has better resistance to false positives and alert prioritization, thus the system is applicable in real-time implementation in operational security settings.

The other important contribution is that of the Explainability and Transparency Intelligence System (X-TIS). The framework allows interpretability issues often related to deep learning-based security systems because it presents both feature-level attribute attribution and graph-based contextual explanations. This openness encourages trust in analysts and can be used to make wise decisions that are essential to actual cybersecurity functions.

Notably, IS-HAITI is not confined to the three models that are currently in place. It has been specifically designed to be extensible and modular enough to add new detection models in the future, to serve other specific threat categories, including insider threats, application-layer threats, IoT-specific threats or new adversarial behavior. With additional models being introduced, as well as trained on various attack models, the system will eventually have a wider threat coverage and an incrementally higher detection accuracy without necessarily a structural change in the basic architecture.

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To validate the model's decision-making, we applied SHAP analysis to high-risk predictions. Fig. 3 confirms that `dst_bytes_mean` (Destination Bytes) is the dominant feature driving detection. This aligns with the "no-leak" feature set design, confirming that the model is learning from volume-based traffic anomalies rather than memorizing specific IP addresses.